# Programming Methodology
# Team Project

Project Guidelines & Overview

Spring 2026

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Machine Intelligence & Data science LAB

# Project Objectives

C++

### Apply C++ OOP Concepts

Classes, inheritance, virtual functions, polymorphism, and etc.

Git

### Team Collaboration via Git

Work in teams of 4 using Git

PM

### Large-Scale Project Planning

Plan, design, and deliver a complete game



# Timeline

| | Date | Milestone | Details |
|---|---|---|---|
| 1 | **4/13** | **Project Release** | Project announcement & guidelines overview |
| 2 | **4/17** | **Team Formation Due** | Finalize teams |
| 3 | **6/1** | **Demo Release** | Publish executable only (no source code) |
| 4 | **6/5** | **Peer Feedback Due** | Provide feedback on other teams' demos |
| 5 | **6/14** | **Final Submission Due** | Incorporate feedback & final submission |
| 6 | **6/15-16** | **Final Presentation** | Presentation |

# Game Explanation

# Game Theme: Farm Village Simulator

**Build a PVE (Player vs. Environment) Farm Simulation game**

Your team will design and implement a text-based (or GUI-based) Farm Village Simulator.

1 **Required C++ Components**

2 **Required Game Features**

# Required C++ Components

- **Class**
- **Inheritance**
- **Virtual Functions & Friend**
- **STL (vector, stack, and etc)**
- **Template**
- **Operator Overloading**
- **Pointer**

**All seven components listed above must be included without exception.**
You may also include additional components covered in class.

# Required Game Features

- **Agriculture: Harvesting & Livestock**
- **Marketplace: Shop & Currency**
- **Construction: Crafting & Building**
- **Level System**

**All four features listed above must be included without exception.**
Keep the concept, but ***feel free*** to implement them.






## Random Event System

Your game **must** include unpredictable events that triggered randomly and affect gameplay.

- Drought
- Pest Infestation
- Flood

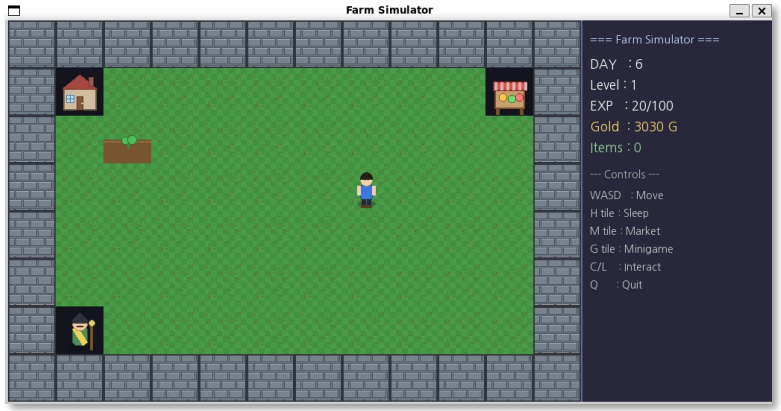
The three events listed above are examples;
**you must include at least one random events systems.**



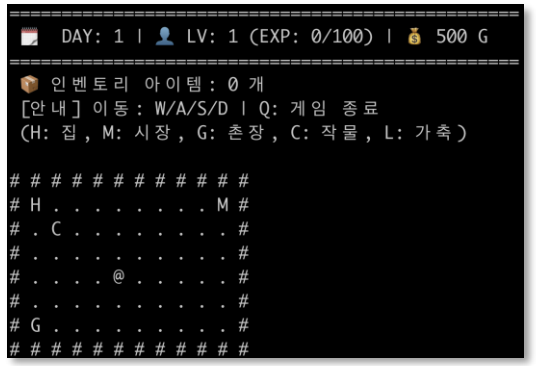
# Optional Features

**You can include any elements that enhance the fun of the game.**
Below is an example. Each team can implement it ***freely***.

Visual **GUI Interface**




w/ GUI




w/o GUI

Fun **Mini Games**

Include embedded mini-games.

# Team Formation and Collaboration

*(submission due: 4/17 / confirmation: 4/18)*

# Team Formation

## 1. Form Your Team

Teams of **4 members**

You may form groups of 1,2, or 4 members. (If you submit as a group of 1 or 2 people, we will group you into a group of 4.)

## 2. Register via Google Form

Submit your team via the provided *Google Form*.

Link: https://docs.google.com/forms/d/1m1x6acnsn2BsjNqymPAUQ7VwGdYc-qp9JnTgBkiJH8A/edit
**Due: 4/17 11:59 pm**

# Git Collaboration

## How to set up a team workspace (Github Organization)

Once your team is confirmed (4/18), please follow these steps to set up your workspace:

1. **GitHub Invitation**

   You will receive an invitation to the GitHub Organization: **2026-Spring-PM**.

   Link: https://github.com/orgs/2026-Spring-PM/repositories

2. **Create New Team**

   Create new team in the ‘Teams’ section following below images. (next slides)

   Then add a member to your team.

3. **Repository Creation**

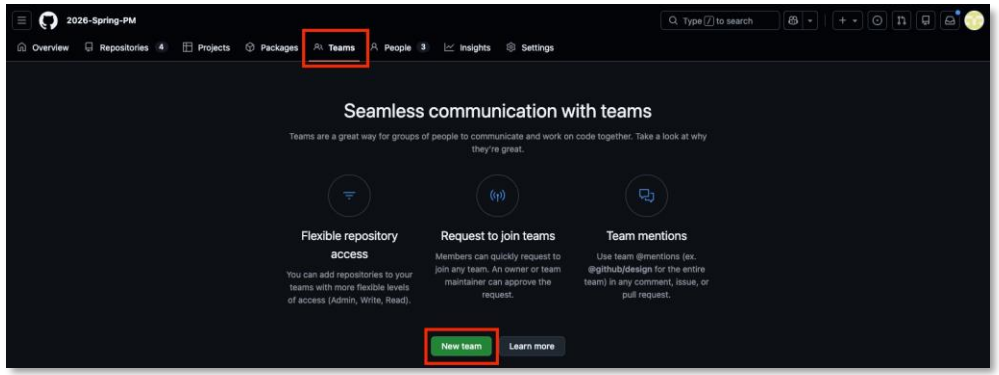
   Within that organization, create a new private repository for your team named **Team_00**.
   (ex. Team_01, Team_02, …)

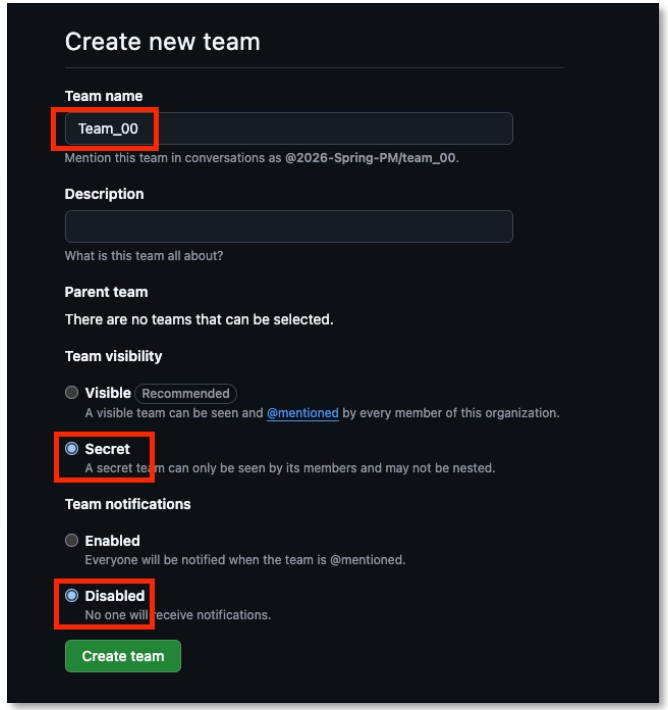
# Git Collaboration

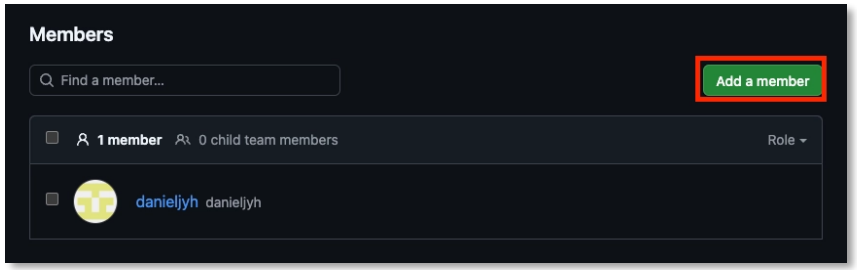
## How to set up a team workspace (Github Organization)

Once your team is confirmed (4/18), please follow these steps to set up your workspace:

### 2. Create New Team



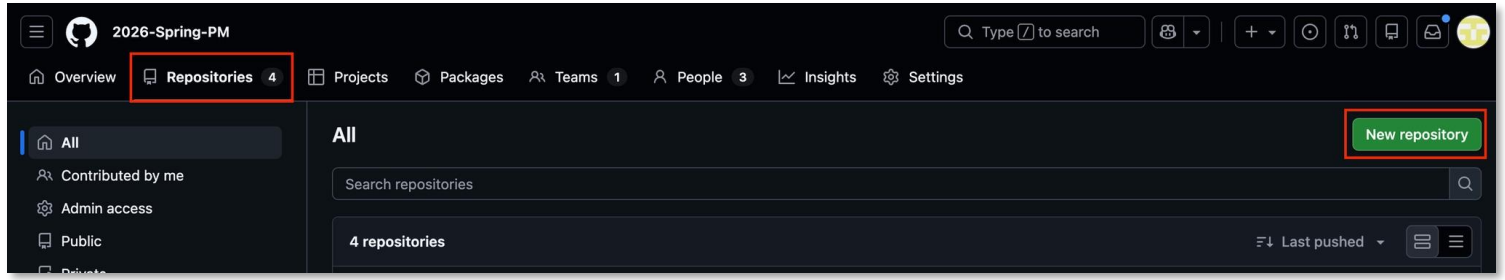



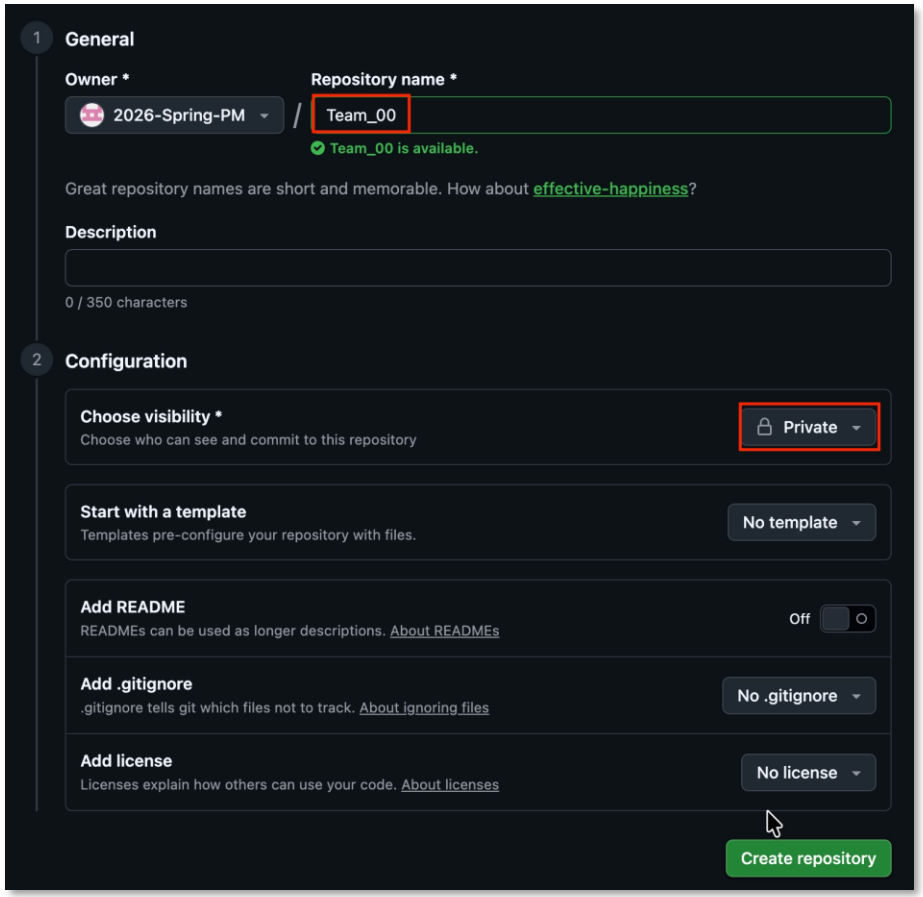
# Git Collaboration

## How to set up a team workspace (Github Organization)

Once your team is confirmed (4/18), please follow these steps to set up your workspace:

### 3. Repository Creation



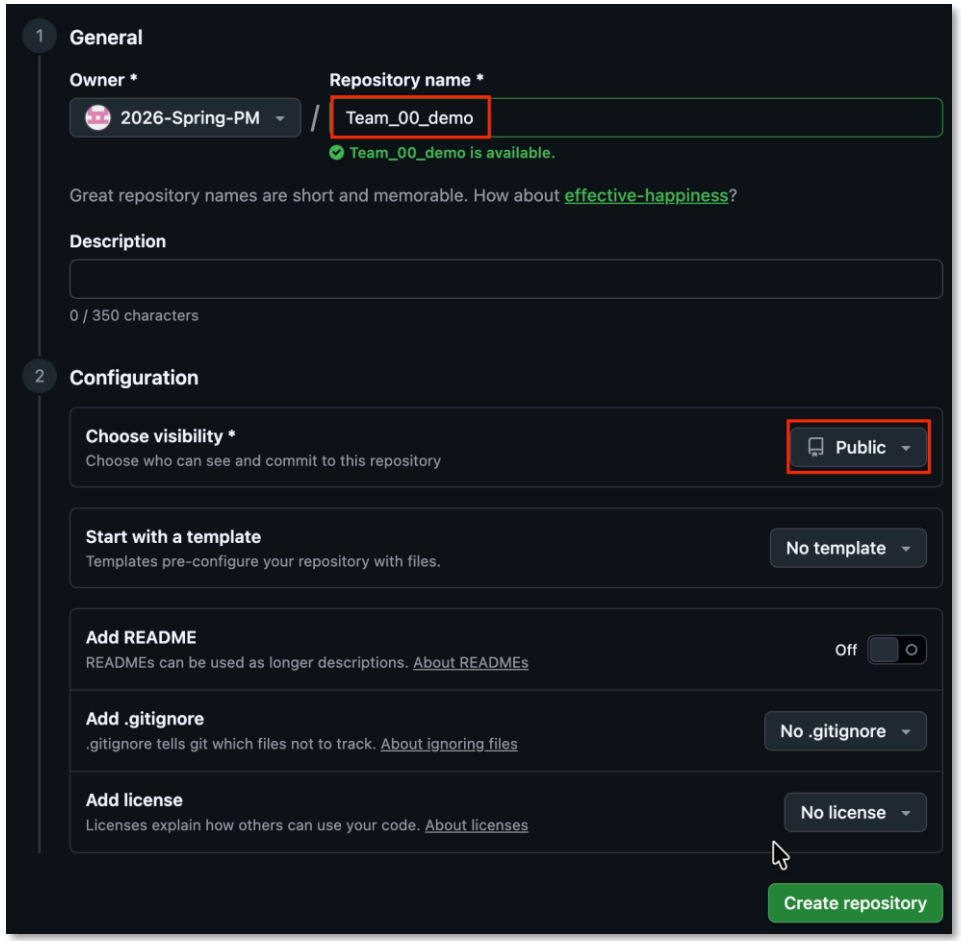
# Mid-term Demo Release

*(submission due: 6/1)*

# Mid-term Demo

**You should create new public repository for demo**

- Repo name: Team_00_demo
- Visability: Public



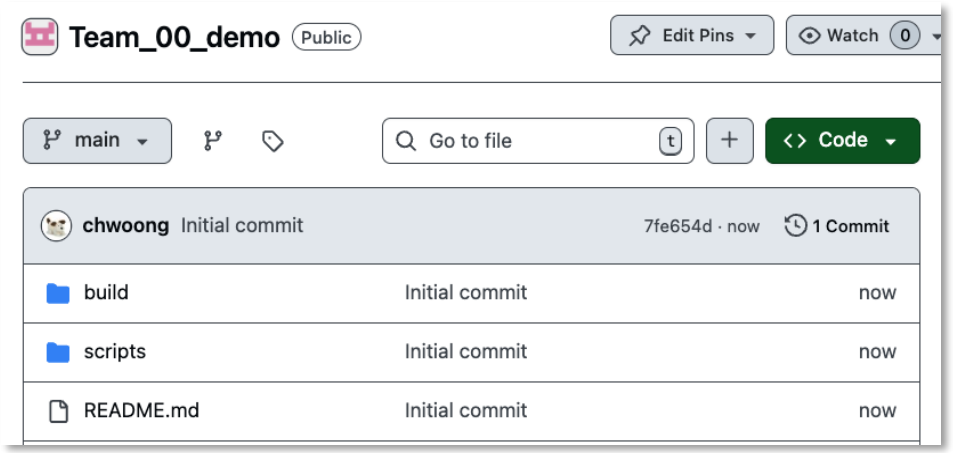
# Mid-term Demo

**For the Mid-term Demo Submission, the code structure within the directory must be as follows.**

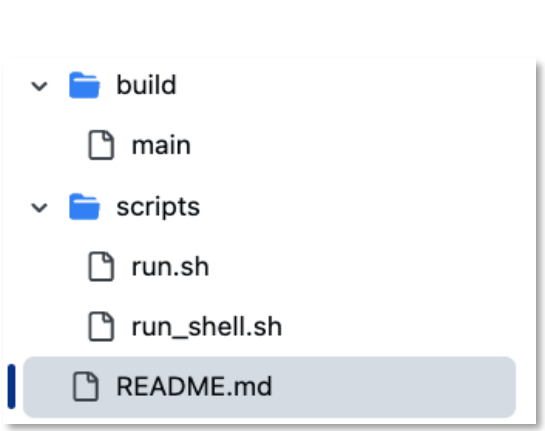
Example:

- https://github.com/2026-Spring-PM/Team_00_demo
- GUI version: https://github.com/2026-Spring-PM/Team_00_demo_gui

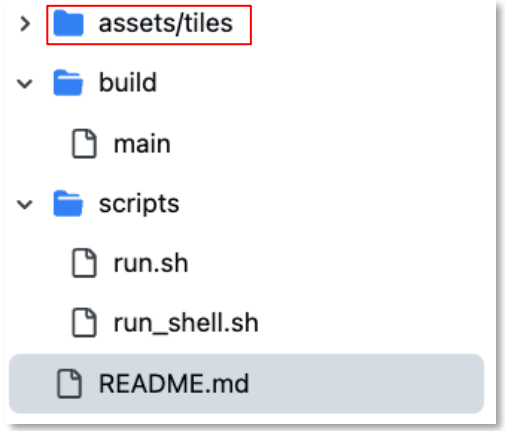
* In this example repository, there is **an explanation of the base Docker image** that can be used for the project.




Repository structure should be ...




or




**Do NOT include source files** in the mid-term demo repository.

# Docker Environment

**For this project, we use Docker!**

**Please refer to Lab2-02** for an in-depth explanation of the Docker environment used in this project.

- **The base Docker image** is in the demo example repository (*chwoong/team_00_project:0.1.0*)
  - This Docker image already includes GCC 13 (g++)

- If this Docker image is **insufficient**, you are allowed to build **a new Docker image** and push it to Docker Hub.
  - In this case, include your team number in the name.

```
docker login
docker tag {image}:{tag} {userid}/{image}:{tag}
# docker tag team1_project:0.1.0 snu12345/team1_project:0.1.0
docker push {userid}/{image}:{tag}
# docker push snu12345/team1_project:0.1.0
```

team number

```
IMAGE_NAME="snu12345/team2_project:0.1.0"

docker run -it --rm \
  --platform linux/amd64 \
  -v $(pwd):/workspace \
  --name farm-sim-container \
  "$IMAGE_NAME" \ \
  /workspace/main
```

# Peer feedback

*(due: 6/5)*

# Peer Feedback on the Mid-term Demo

- **Example types of Feedback for Other Teams**
  - **Bug** : Errors or abnormal behavior found in the game
  - **Enhancement** : Features or improvements that would make the game better
  - **Encouragement** : Elements you liked or found impressive
  - **Error**: If the game does not run
    - Clearly describe where the issue occurred (w/ screenshots or explanations)

  * You must provide a reason that supports your feedback

- **How to Submit Feedback**
  - Use Github Issues
  - Each student **must** give feedback to **2 teams**

  * Details about submission and the assigned team numbers will be announced later.

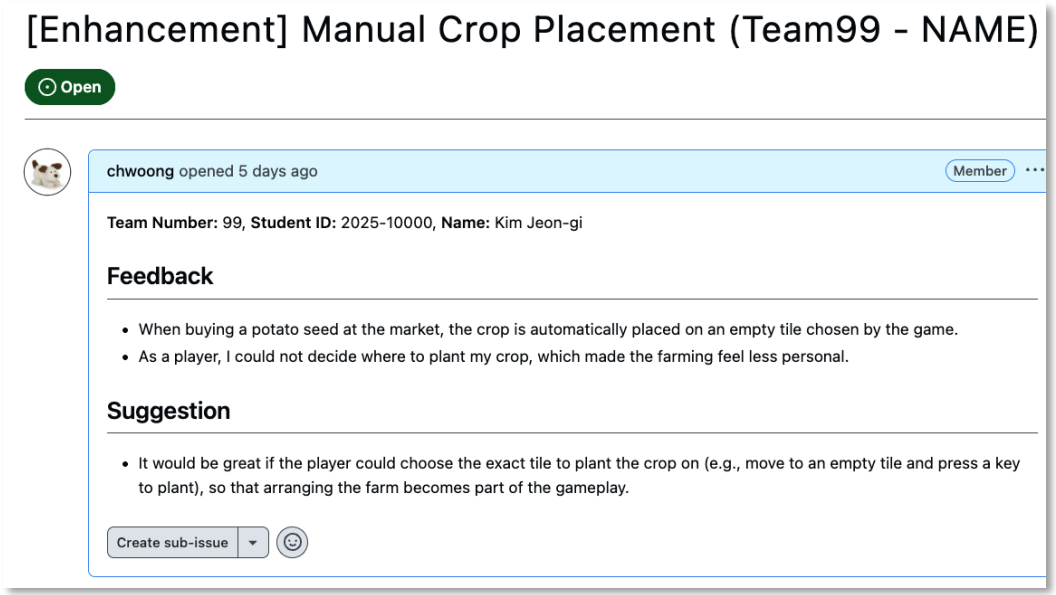
# Peer Feedback on the Mid-term Demo

Example:

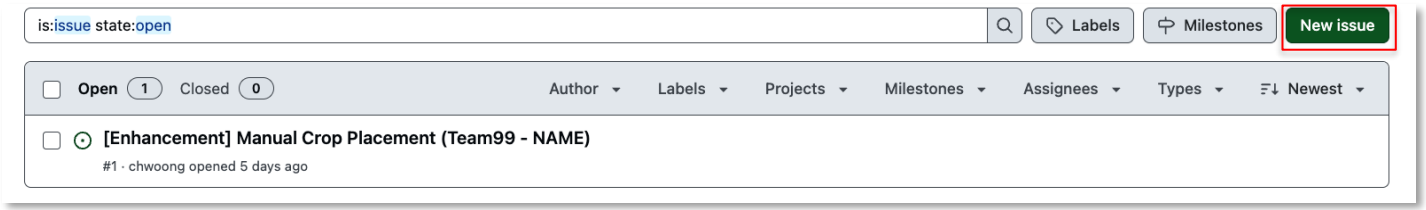
- https://github.com/2026-Spring-PM/Team_00_demo/issues
- https://github.com/2026-Spring-PM/Team_00_demo_gui/issues

Title format:

- **[FEEDBACK-TYPE]** Detailed title... **(Team00 – NAME)**




**feedback example**



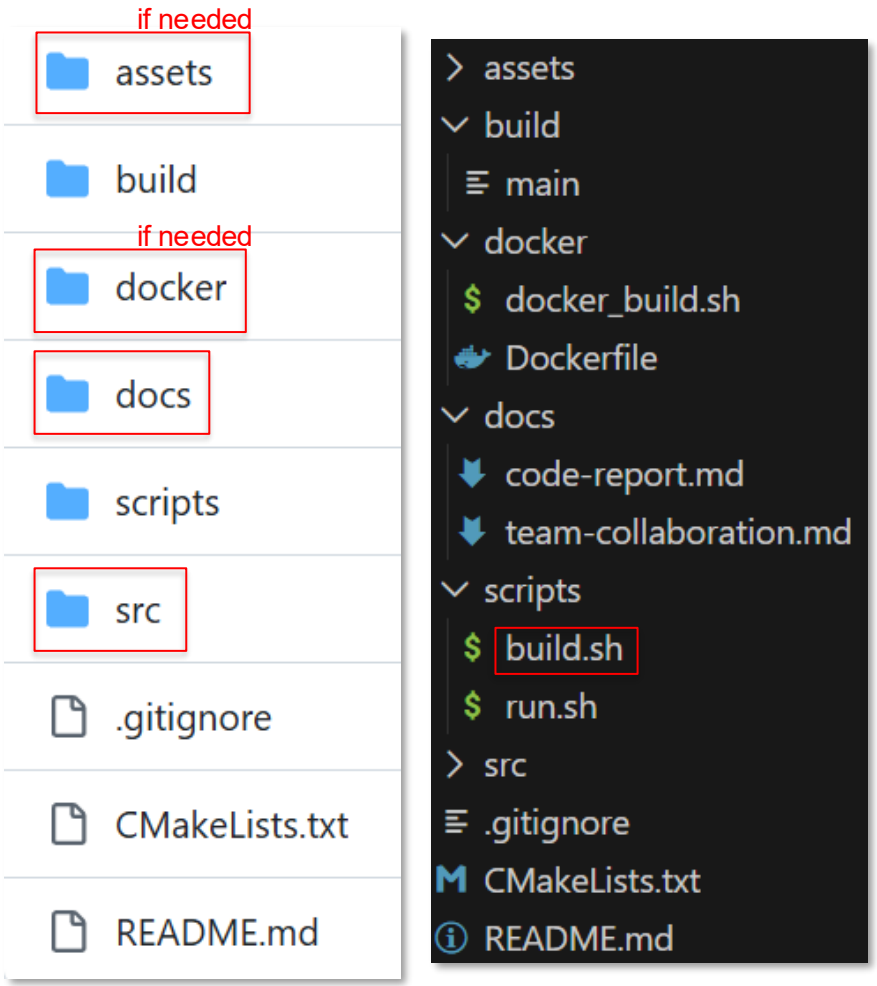

**Github Issues example**

# Final Submission

*(submission due: 6/14)*

# Final Submission

**For the Final Submission, the code structure within the directory must be as follows.**




1. **scripts/build.sh**
   - For the compilation process or CMake
2. **Full Source Code** (src/*.cpp, src/*.h)
   - All C++ source and header files must be included
3. **Final Reports** (docs/*.md)

→ These are the key differences from the mid-term demo repo.

# Submission File Contents Requirements

This document includes descriptions of the files required for both the mid-term demo and the final submission.

- **README.md**
  1. How to run the game
     - Write the instructions step by step in enough detail so that others can run the game without errors.
  2. Brief introduction and core features
     - Include a short introduction to the game and briefly explain its main features

- **scripts/run.sh**
  - A shell script for running Docker and launching the game
  - When bash run.sh is executed, Docker should start first, and the game should launch immediately inside the container

- **scripts/build.sh (final submission only)**
  - A bash script that includes the compilation process or CMake
  - When build.sh is executed, a main executable file should be created in the build folder
  - The build instructions must also be included in README.md

# Submission File Contents Requirements

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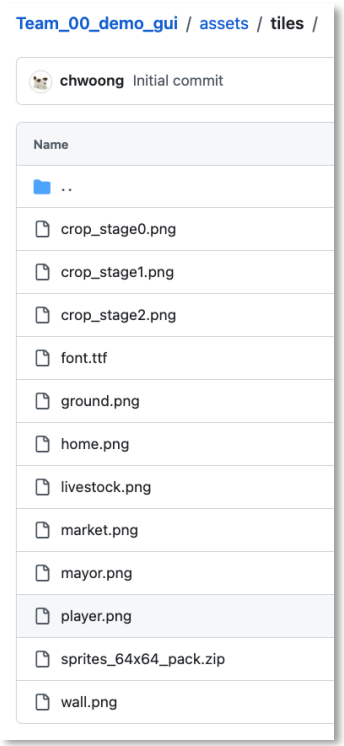
- **docker/* (if needed)**
  - Not required if you use the provided default Docker image (`chwoong/team_00_project:0.1.0`)
  - Include:
    - Dockerfile
    - docker_build.sh : running this script should build the Docker image

- **assets/* (if needed)**
  - Assets required to run the game (images, fonts)
  - The folder structure is flexible

- **.gitignore**
  - CMake builds generate various files in addition to main
  - Use .gitignore so that unnecessary generated files are excluded from git commits

```
# CMake generated files
CMakeFiles/
CMakeCache.txt
cmake_install.cmake
Makefile
install_manifest.txt
compile_commands.json
```

.gitignore example




assets/* example

# Submission File Contents Requirements

This document includes descriptions of the files required for both the mid-term demo and the final submission.

- **docs/*.md (final submission only)**
  - Write the report in Markdown
  - Include only essential and core content
    * Writing more does not lead to a higher score
  - Length limit: ~A4 1page

1. **docs/code-report.md**
   Include the following:

   a) The overall module structure of the code

   b) Explanation of the code implementing the required system components

   c) Explanation of where and how required C++ concepts were used

   d) AI utilization (only if generative AI was used)

   * For AI utilization, you may write freely, but please include:
     - What purpose AI was used for
     - What changes were made to the code using AI
     - If the generated code had errors, how those errors were identified and fixed

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    * Writing more does not lead to a higher score
  - Length limit: ~A4 1page

2. **docs/team-collaboration.md**
   Include the following:

   a) What each team member contributed (written individually)
      - Up to 3 lines per person

   b) Explanation of your personal Top 3 best Git commits

   c) Attach the Git graph of your repository
      - Briefly explain which team member worked on what in the important commits
      - You do not need to attach the entire Git log, only include the important parts

# Final presentation

*(6/15 & 6/16)*

# Final Presentation: Peer Evaluation

**Presentation day : 6/15 Monday, 6/16 Tuesday**

- Schedule will be announced later.
- Presenter(s): flexible — one or all members

- **Presentation Structure (approx. 8 min, tentative)**
  - Game design
  - Game demo (recording and screenshots)
  - Peer feedback received → improvements made
  - Include AI utilization (e.g., useful insights or tips to share)

**After each team presents, all other teams evaluate them.**

1. **Creativity**
   - Does it include unique or creative gameplay elements?
2. **Playability**
   - Would you actually want to play this game? (fun or engaging)
3. **Presentation**
   - Is the presentation well presented?

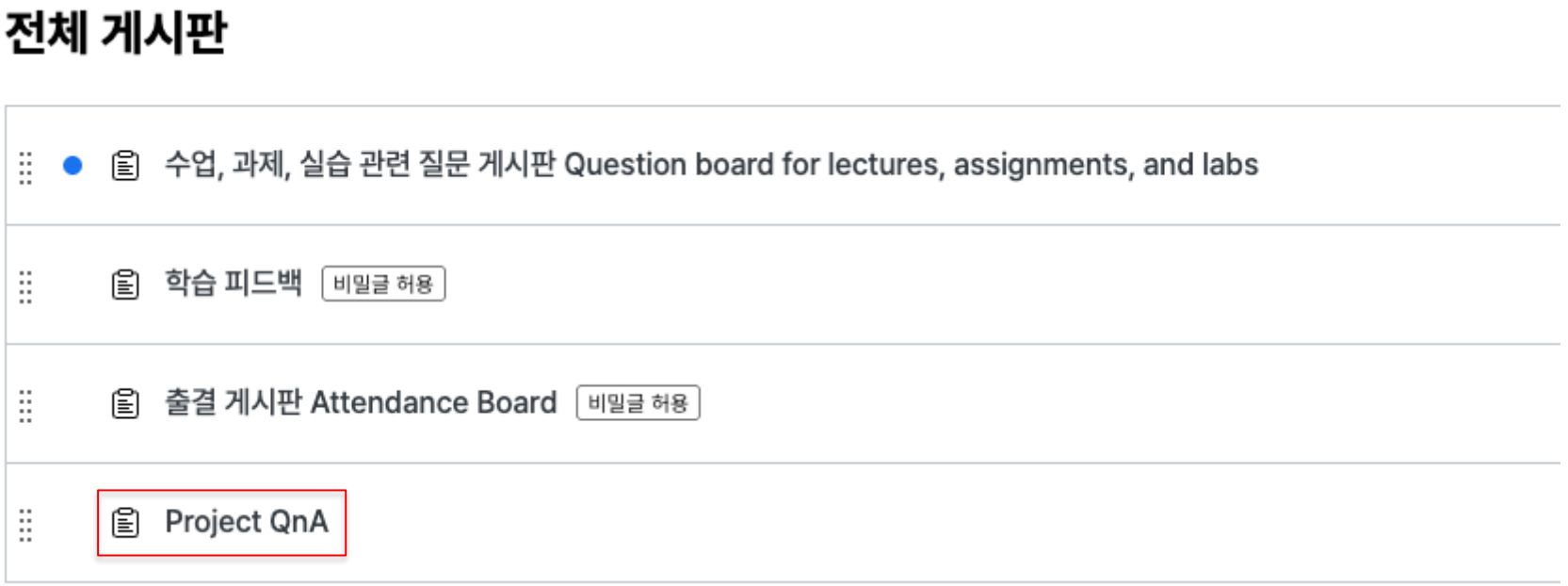
**Top N Ranked Teams → Bonus Points!**

# Grading Criteria

# Grading Criteria

- **Team Formation** +5
- **Implementation (Mid-term Demo)** +20
- **Peer Feedback (Mid-term Demo)** +20
- **Implementation (Final Submission)** +20
- **Report (Final Submission)** +15
- **Final presentation** +20

# QnA / Tips


전체 게시판
수업, 과제, 실습 관련 질문 게시판 Question board for lectures, assignments, and labs
학습 피드백
비밀글 허용
출결 게시판 Attendance Board
비밀글 허용
Project QnA

# For Windows users...

For Windows users, please use Windows Terminal to avoid rendering issues (ex. Emoji)

